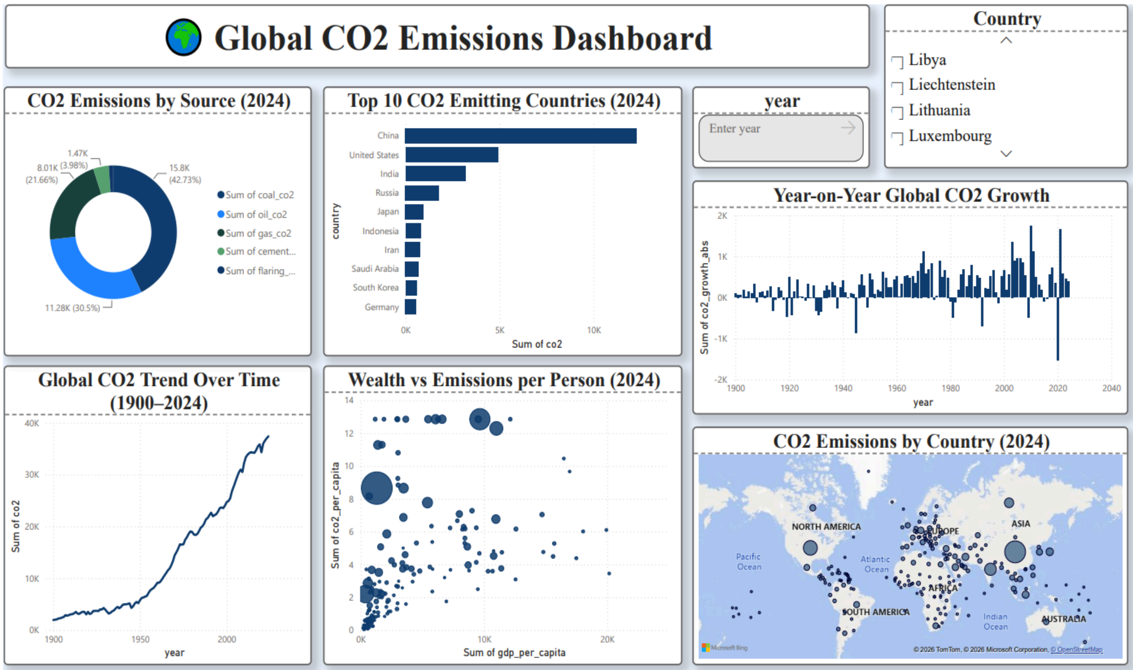
Global CO2 Emissions Analysis Report

1. Dashboard



2. Findings

Global CO2 emissions have increased dramatically from 1900 to 2024, rising from nearly zero to over 37,000 million tonnes. Growth accelerated sharply after 1950 due to rapid industrialization. Coal remains the largest source at 42.73%, followed by oil at 30.5% and gas at 21.66%. China leads global emissions in 2024, followed by the USA and India. Brief dips occurred during the 2008 financial crisis and 2020 COVID-19 pandemic but emissions quickly recovered.

3. Key Insights

- ❖ Coal accounts for 42.73% of all CO_2 emissions — the single largest source.
- ❖ China emits nearly double the CO_2 of the second largest emitter USA in 2024.
- ❖ Global CO_2 has grown exponentially since 1950 showing no signs of slowing.
- ❖ Wealthier nations tend to have higher CO_2 emissions per person.
- ❖ 2020 shows the largest single year drop in CO_2 growth due to COVID-19 lockdowns.

4. Recommendations

- ★ Phase Out Coal — Replacing coal with renewables would have the highest impact since coal contributes 42.73% of emissions.
- ★ Target Top 3 Emitters — China, USA and India together produce over 50% of global CO_2. International agreements must prioritize these nations.
- ★ Clean Technology Transfer — Help developing nations adopt solar and wind energy to skip coal-based industrialization.
- ★ Learn from COVID Dip — The 2020 drop proves emissions can fall rapidly. Policies promoting remote work can help maintain lower levels.

5. Conclusion

This analysis reveals that global CO_2 emissions have grown dramatically over the past century, driven primarily by coal consumption and industrialization in major economies. Immediate action targeting the top emitting nations and phasing out coal energy is critical to reversing this trend and addressing climate change.